

Capstone Project on Multi Modal

Project Title

AI Multimodal Smart Knowledge Assistant

1. Problem Statement

Develop an **AI Multimodal Smart Knowledge Assistant** that supports multiple input and output modalities.

The assistant should:

- Accept text queries.
- Accept speech input.
- Accept image uploads.
- Generate captions for uploaded images.
- Retrieve relevant information from a knowledge base.
- Generate intelligent responses.
- Convert responses into speech.
- Provide an easy-to-use Gradio interface.

2. Functional Requirements

Module 1 – Text and Speech

The application shall support:

a) Text Input

Users can type questions.

b) Speech Input

Users can speak a question.

Speech shall be converted into text using Speech-to-Text.

c) Text Response

The assistant shall answer the question in text.

d) Speech Response

The generated answer shall also be converted into speech using Text-to-Speech.

Module 2 – Vision

The application shall support:

Image Upload

Users upload an image.

Image Captioning

Generate an appropriate caption describing the uploaded

image. **Image Generation**

Generate an image from a text prompt.

Module 3 – Multimodal RAG

The assistant should retrieve relevant information from a small knowledge

base. The knowledge base may contain information related to:

- College
- Tourism
- Healthcare
- Agriculture
- Library
- Museums
- Historical monuments

6. Technologies* (Note*: will updated as per current trend)

Component	Suggested Technology
User Interface	Gradio
Speech-to-Text	Whisper
Text-to-Speech	gTTS
Image Captioning	Llama Vision Model
Image Generation	DALL·E
Retrieval	LangChain
LLM	Mixtral / Llama